Lab 6: Introducing Classification

Objectives:

- To gain hands-on experience classifying small dataset
- To implement concepts related to Decision Tree classifier (i.e. Entropy, Information Gain), along with using existing libraries.

WISIT SUWANNAO 67070501042

In [1]:
```python
# Run this cell if you use Colab
from google.colab import drive
drive.mount('/content/drive')
```

```
---------------------------------------------------------------------------
ModuleNotFoundError                       Traceback (most recent call last)
Cell In[1], line 2
      1 # Run this cell if you use Colab
----> 2 from google.colab import drive
      3 drive.mount('/content/drive')

ModuleNotFoundError: No module named 'google'
```

Code it yourself

In [1]:
```python
import pandas as pd

# Read the data
df = pd.read_csv('toy_data.csv')
df
```

Out[1]:

	age	income	student	credit rating	buys computer
0	<=30	high	no	fair	no
1	<=30	high	no	excellent	no
2	31-40	high	no	fair	yes
3	>40	medium	no	fair	yes
4	>40	low	yes	fair	yes
5	>40	low	yes	excellent	no
6	31-40	low	yes	excellent	yes
7	<=30	medium	no	fair	no
8	<=30	low	yes	fair	yes
9	>40	medium	yes	fair	yes
10	<=30	medium	yes	excellent	yes
11	31-40	medium	no	excellent	yes
12	31-40	high	yes	fair	yes
13	>40	medium	no	excellent	no

In [2]:
```python
print(df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 14 entries, 0 to 13
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   age            14 non-null     str
 1   income         14 non-null     str
 2   student        14 non-null     str
 3   credit rating  14 non-null     str
 4   buys computer  14 non-null     str
dtypes: str(5)
memory usage: 692.0 bytes
None
```

TODO: Write functions to compute Gain and Entropy, as discussed in the lecture.

In [3]:
```python
# Write your code here
import numpy as np

def entropy(target_col):
    """
    Calculate the entropy of a dataset.
    The only parameter of this function is the target_col parameter which specifies the target column
    """
    elements, counts = np.unique(target_col, return_counts=True)
    entropy_val = 0
    sum_counts = np.sum(counts)
    for i in range(len(elements)):
        p_i = counts[i] / sum_counts
        entropy_val += -p_i * np.log2(p_i)
    return entropy_val

def information_gain(data, split_attribute_name, target_name="buys computer"):
    """
    Calculate the information gain of a dataset.
    """
    # Calculate the entropy of the total dataset
    total_entropy = entropy(data[target_name])

    # Calculate the values and the counts for the split attribute
    vals, counts = np.unique(data[split_attribute_name], return_counts=True)

    # Calculate the weighted entropy
    weighted_entropy = 0
    sum_counts = np.sum(counts)
    for i in range(len(vals)):
        p_v = counts[i] / sum_counts

        # Filter the data for the specific value
        filtered_data = data[data[split_attribute_name] == vals[i]]

        # call the entropy function for the target column of the filtered data
        weighted_entropy += p_v * entropy(filtered_data[target_name])

    # Calculate the information gain
    information_gain_val = total_entropy - weighted_entropy
    return information_gain_val

# Calculate Information Gain for each feature
features = ['age', 'income', 'student', 'credit rating']
print(f"Entropy of dataset (buys computer): {entropy(df['buys computer']):.4f}")

print("\nInformation Gain for each feature:")
for feature in features:
    ig = information_gain(df, feature)
    print(f"{feature}: {ig:.4f}")
```

```
Entropy of dataset (buys computer): 0.9403

Information Gain for each feature:
age: 0.2467
income: 0.0292
student: 0.1518
credit rating: 0.0481
```

Using Libraries

Now that you know how to compute these values by yourselfs, now let's use some libraries.

Steps:

- Split the Data → Divide dataset into training (80%) and testing (20%).
- Train the Model → Fit a Decision Tree using the training data.
- Test the Model → Use the trained model to predict on test data.
- Evaluate Performance → Compare predictions with actual values (e.g., Accuracy Score).

Prepare features and labels.

In [4]:
```python
# Features
features = df.drop('buys computer', axis=1)
features

# Alternatively, you can use this:
# features = df.iloc[:, :-1]
```

Out[4]:

	age	income	student	credit rating
0	<=30	high	no	fair
1	<=30	high	no	excellent
2	31-40	high	no	fair
3	>40	medium	no	fair
4	>40	low	yes	fair
5	>40	low	yes	excellent
6	31-40	low	yes	excellent
7	<=30	medium	no	fair
8	<=30	low	yes	fair
9	>40	medium	yes	fair
10	<=30	medium	yes	excellent
11	31-40	medium	no	excellent
12	31-40	high	yes	fair
13	>40	medium	no	excellent

In [5]:
```python
# Labels (or Target)
labels = df['buys computer']
labels

# # Alternatively, you can use this:
# labels = df.iloc[:, [-1]]
```

Out[5]:
```
0      no
1      no
2     yes
3     yes
4     yes
5      no
6     yes
7      no
8     yes
9     yes
10    yes
11    yes
12    yes
13     no
Name: buys computer, dtype: str
```

In [6]:
```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree

# 1. Load the dataset
X = features.values  # Features
y = labels.values  # Target labels

# 2. Split the dataset into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Create and train a Decision Tree model with entropy criterion
clf = DecisionTreeClassifier(criterion='entropy', max_depth=3, random_state=42)
clf.fit(X_train, y_train)
```

```
Matplotlib is building the font cache; this may take a moment.
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[6], line 15
     13 # 3. Create and train a Decision Tree model with entropy criterion
     14 clf = DecisionTreeClassifier(criterion='entropy', max_depth=3, random_state=42)
---> 15 clf.fit(X_train, y_train)

File c:\Users\Acer\Downloads\CPE232\.venv\Lib\site-packages\sklearn\base.py:1336, in _fit_context.<locals>.decorator.<locals>.wrapper(estimator, *args, **kwar
gs)
   1329     estimator._validate_params()
   1331 with config_context(
   1332     skip_parameter_validation=(
   1333         prefer_skip_nested_validation or global_skip_validation
   1334     )
   1335 ):
-> 1336     return fit_method(estimator, *args, **kwargs)

File c:\Users\Acer\Downloads\CPE232\.venv\Lib\site-packages\sklearn\tree\_classes.py:1027, in DecisionTreeClassifier.fit(self, X, y, sample_weight, check_inpu
t)
    996 @_fit_context(prefer_skip_nested_validation=True)
    997 def fit(self, X, y, sample_weight=None, check_input=True):
    998     """Build a decision tree classifier from the training set (X, y).
    999 
   1000     Parameters
   (...)
   1024         Fitted estimator.
   1025     """
-> 1027     super()._fit(
   1028         X,
   1029         y,
   1030         sample_weight=sample_weight,
   1031         check_input=check_input,
   1032     )
   1033     return self

File c:\Users\Acer\Downloads\CPE232\.venv\Lib\site-packages\sklearn\tree\_classes.py:255, in BaseDecisionTree._fit(self, X, y, sample_weight, check_input, mis
sing_values_in_feature_mask)
    251 check_X_params = dict(
    252     dtype=DTYPE, accept_sparse="csc", ensure_all_finite=False
    253 )
    254 check_y_params = dict(ensure_2d=False, dtype=None)
--> 255 X, y = validate_data(
    256     self, X, y, validate_separately=(check_X_params, check_y_params)
    257 )
    259 missing_values_in_feature_mask = (
    260     self._compute_missing_values_in_feature_mask(X)
    261 )
    262 if issparse(X):

File c:\Users\Acer\Downloads\CPE232\.venv\Lib\site-packages\sklearn\utils\validation.py:2914, in validate_data(_estimator, X, y, reset, validate_separately, s
kip_check_array, **check_params)
   2912 if "estimator" not in check_X_params:
   2913     check_X_params = {**default_check_params, **check_X_params}
-> 2914 X = check_array(X, input_name="X", **check_X_params)
   2915 if "estimator" not in check_y_params:
   2916     check_y_params = {**default_check_params, **check_y_params}

File c:\Users\Acer\Downloads\CPE232\.venv\Lib\site-packages\sklearn\utils\validation.py:1022, in check_array(array, accept_sparse, accept_large_sparse, dtype,
order, copy, force_writeable, ensure_all_finite, ensure_non_negative, ensure_2d, allow_nd, ensure_min_samples, ensure_min_features, estimator, input_name)
   1020     array = xp.astype(array, dtype, copy=False)
   1021 else:
-> 1022     array = _asarray_with_order(array, order=order, dtype=dtype, xp=xp)
   1023 except ComplexWarning as complex_warning:
   1024     raise ValueError(
   1025         "Complex data not supported\n{}\n".format(array)
   1026     ) from complex_warning

File c:\Users\Acer\Downloads\CPE232\.venv\Lib\site-packages\sklearn\utils\_array_api.py:878, in _asarray_with_order(array, dtype, order, copy, xp, device)
    876     array = numpy.array(array, order=order, dtype=dtype)
    877 else:
--> 878     array = numpy.asarray(array, order=order, dtype=dtype)
    880 # At this point array is a NumPy ndarray. We convert it to an array
    881 # container that is consistent with the input's namespace.
    882 return xp.asarray(array)

ValueError: could not convert string to float: '31-40'
```

There's an error:

ValueError: could not convert string to float: '31-40'

In [7]:
```python
from sklearn.preprocessing import LabelEncoder

# Initialize LabelEncoder
label_encoder = LabelEncoder()

# Apply Label Encoding for all categorical columns
df['age'] = label_encoder.fit_transform(df['age'])
df['income'] = label_encoder.fit_transform(df['income'])
df['student'] = label_encoder.fit_transform(df['student'])
df['credit rating'] = label_encoder.fit_transform(df['credit rating'])
df['buys computer'] = label_encoder.fit_transform(df['buys computer'])

# Display the encoded DataFrame
print(df)
```

```
    age  income  student  credit rating  buys computer
0     1       0        0              1              0
1     1       0        0              0              0
2     0       0        0              1              1
3     2       2        0              1              1
4     2       1        1              1              1
5     2       1        1              0              0
6     0       1        1              0              1
7     1       2        0              1              0
8     1       1        1              1              1
9     2       2        1              1              1
10    1       2        1              0              1
11    0       2        0              0              1
12    0       0        1              1              1
13    2       2        0              0              0
```

Let's check out an updated dataframe.

In [8]:
```python
df
```

Out[8]:

	age	income	student	credit rating	buys computer
0	1	0	0	1	0
1	1	0	0	0	0
2	0	0	0	1	1
3	2	2	0	1	1
4	2	1	1	1	1
5	2	1	1	0	0
6	0	1	1	0	1
7	1	2	0	1	0
8	1	1	1	1	1
9	2	2	1	1	1
10	1	2	1	0	1
11	0	2	0	0	1
12	0	0	1	1	1
13	2	2	0	0	0

In [9]:
```python
X = df.drop('buys computer', axis=1)  # Features
y = df['buys computer']  # Target
```

Let's continue where we left off!

In [10]:
```python
# 2. Split the dataset into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Create and train a Decision Tree model with entropy criterion
clf = DecisionTreeClassifier(criterion='entropy', max_depth=3, random_state=42)
clf.fit(X_train, y_train)
```

Out[10]:

▸ DecisionTreeClassifier ? i

▸ Parameters

In [11]:
```python
print(X_train.shape)
print(X_test.shape)
```

```
(11, 4)
(3, 4)
```

Now we're going build the Decision Tree Classifier

In [12]:
```python
from sklearn.tree import DecisionTreeClassifier

# Initialize the Decision Tree classifier
clf = DecisionTreeClassifier(criterion='entropy', random_state=42)  # Using 'entropy' as the criterion

# Train the model
clf.fit(X_train, y_train)

# Predict on the test set
y_pred = clf.predict(X_test)
```

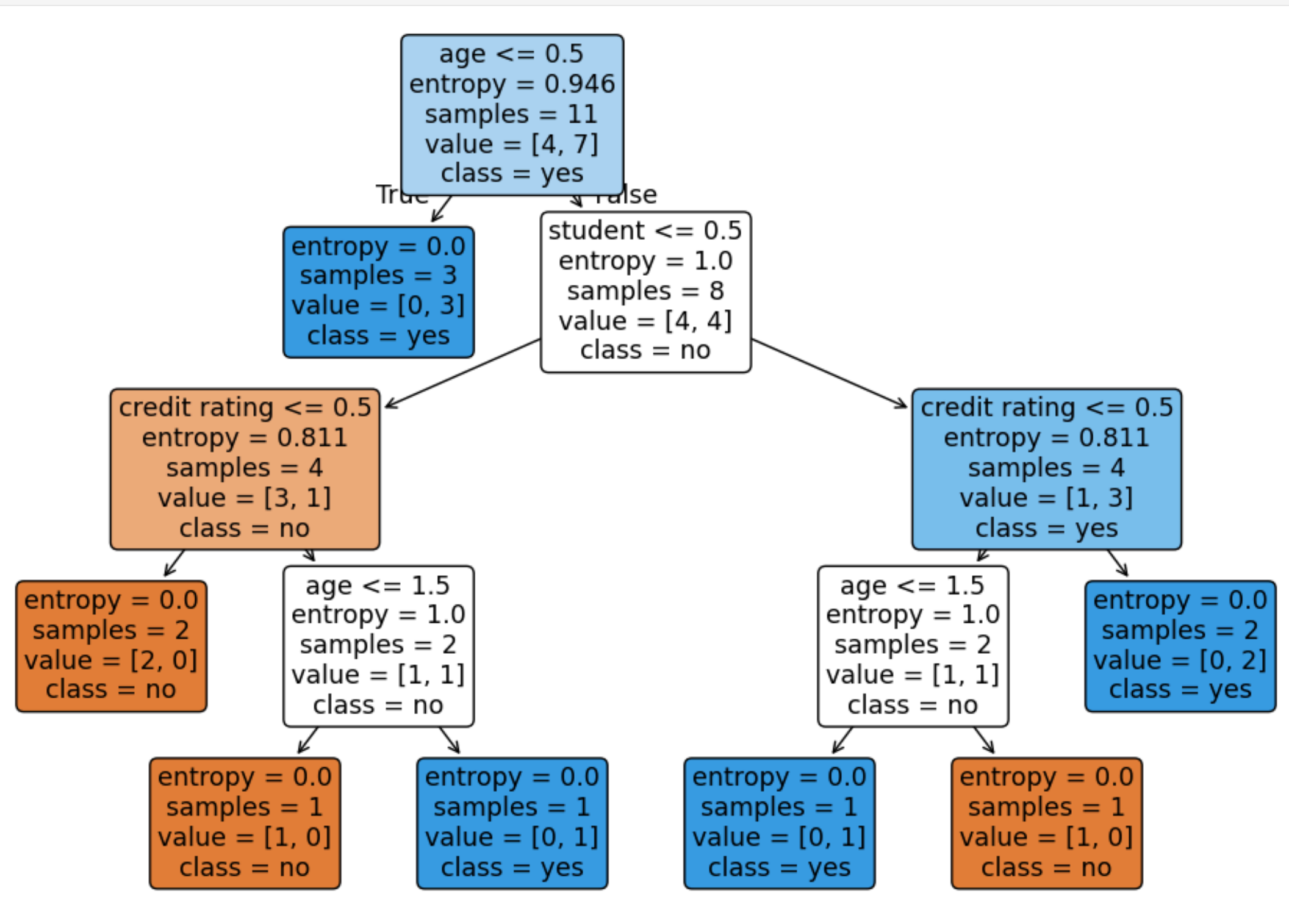
And evaluate our model.

In [13]:
```python
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.2f}")

# Classification report
print("Classification Report:")
print(classification_report(y_test, y_pred))

# Confusion Matrix
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
Accuracy: 1.00
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         1
           1       1.00      1.00      1.00         2

    accuracy                           1.00         3
   macro avg       1.00      1.00      1.00         3
weighted avg       1.00      1.00      1.00         3

Confusion Matrix:
[[1 0]
 [0 2]]
```

And visualize our tree!

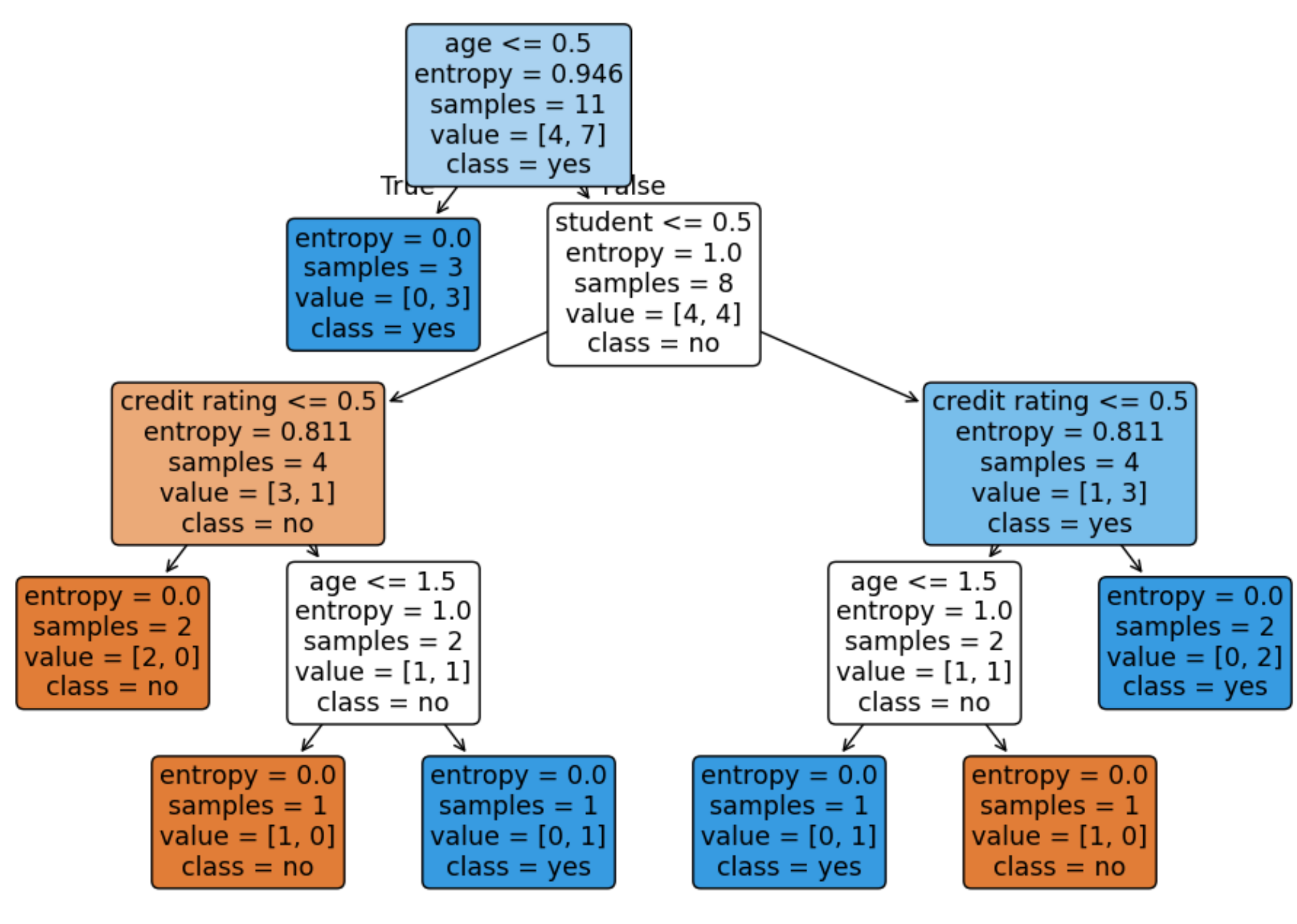
In [14]:
```python
import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

# Plot the decision tree
plt.figure(figsize=(12, 8))
plot_tree(clf, filled=True, feature_names=X.columns, class_names=['no', 'yes'], rounded=True)
plt.show()
```


age <= 0.5
entropy = 0.946
samples = 11
value = [4, 7]
class = yes
True
False
entropy = 0.0
samples = 3
value = [0, 3]
class = yes
student <= 0.5
entropy = 1.0
samples = 8
value = [4, 4]
class = no
credit rating <= 0.5
entropy = 0.811
samples = 4
value = [3, 1]
class = no
credit rating <= 0.5
entropy = 0.811
samples = 4
value = [1, 3]
class = yes
entropy = 0.0
samples = 2
value = [2, 0]
class = no
age <= 1.5
entropy = 1.0
samples = 2
value = [1, 1]
class = no
age <= 1.5
entropy = 1.0
samples = 2
value = [1, 1]
class = no
entropy = 0.0
samples = 2
value = [0, 2]
class = yes
entropy = 0.0
samples = 1
value = [1, 0]
class = no
entropy = 0.0
samples = 1
value = [0, 1]
class = yes
entropy = 0.0
samples = 1
value = [0, 1]
class = yes
entropy = 0.0
samples = 1
value = [1, 0]
class = no

Put them all together.

In [15]:
```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

# data
data = pd.read_csv('toy_data.csv')

df = pd.DataFrame(data)

# Encode categorical columns using LabelEncoder
label_encoder = LabelEncoder()
df['age'] = label_encoder.fit_transform(df['age'])
df['income'] = label_encoder.fit_transform(df['income'])
df['student'] = label_encoder.fit_transform(df['student'])
df['credit rating'] = label_encoder.fit_transform(df['credit rating'])
df['buys computer'] = label_encoder.fit_transform(df['buys computer'])

# Separate features (X) and target (y)
X = df.drop('buys computer', axis=1)
y = df['buys computer']

# Split the dataset into training and test sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize the Decision Tree classifier
clf = DecisionTreeClassifier(criterion='entropy', random_state=42)

# Train the model
clf.fit(X_train, y_train)

# Predict on the test set
y_pred = clf.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.2f}")

# Classification report
print("Classification Report:")
print(classification_report(y_test, y_pred))

# Confusion Matrix
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Plot the decision tree
plt.figure(figsize=(12, 8))
plot_tree(clf, filled=True, feature_names=X.columns, class_names=['no', 'yes'], rounded=True)
plt.show()
```

```
Accuracy: 1.00
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         1
           1       1.00      1.00      1.00         2

    accuracy                           1.00         3
   macro avg       1.00      1.00      1.00         3
weighted avg       1.00      1.00      1.00         3

Confusion Matrix:
[[1 0]
 [0 2]]
```


age <= 0.5
entropy = 0.946
samples = 11
value = [4, 7]
class = yes
True
False
entropy = 0.0
samples = 3
value = [0, 3]
class = yes
student <= 0.5
entropy = 1.0
samples = 8
value = [4, 4]
class = no
credit rating <= 0.5
entropy = 0.811
samples = 4
value = [3, 1]
class = no
credit rating <= 0.5
entropy = 0.811
samples = 4
value = [1, 3]
class = yes
entropy = 0.0
samples = 2
value = [2, 0]
class = no
age <= 1.5
entropy = 1.0
samples = 2
value = [1, 1]
class = no
age <= 1.5
entropy = 1.0
samples = 2
value = [1, 1]
class = no
entropy = 0.0
samples = 2
value = [0, 2]
class = yes
entropy = 0.0
samples = 1
value = [1, 0]
class = no
entropy = 0.0
samples = 1
value = [0, 1]
class = yes
entropy = 0.0
samples = 1
value = [0, 1]
class = yes
entropy = 0.0
samples = 1
value = [1, 0]
class = no

Is the output tree the same as what you calculated yourself? Explain in your own words why they are the same or different.

Ans: The output tree is likely **different** from a manually calculated one (assuming standard ID3 algorithm) for the following reasons:

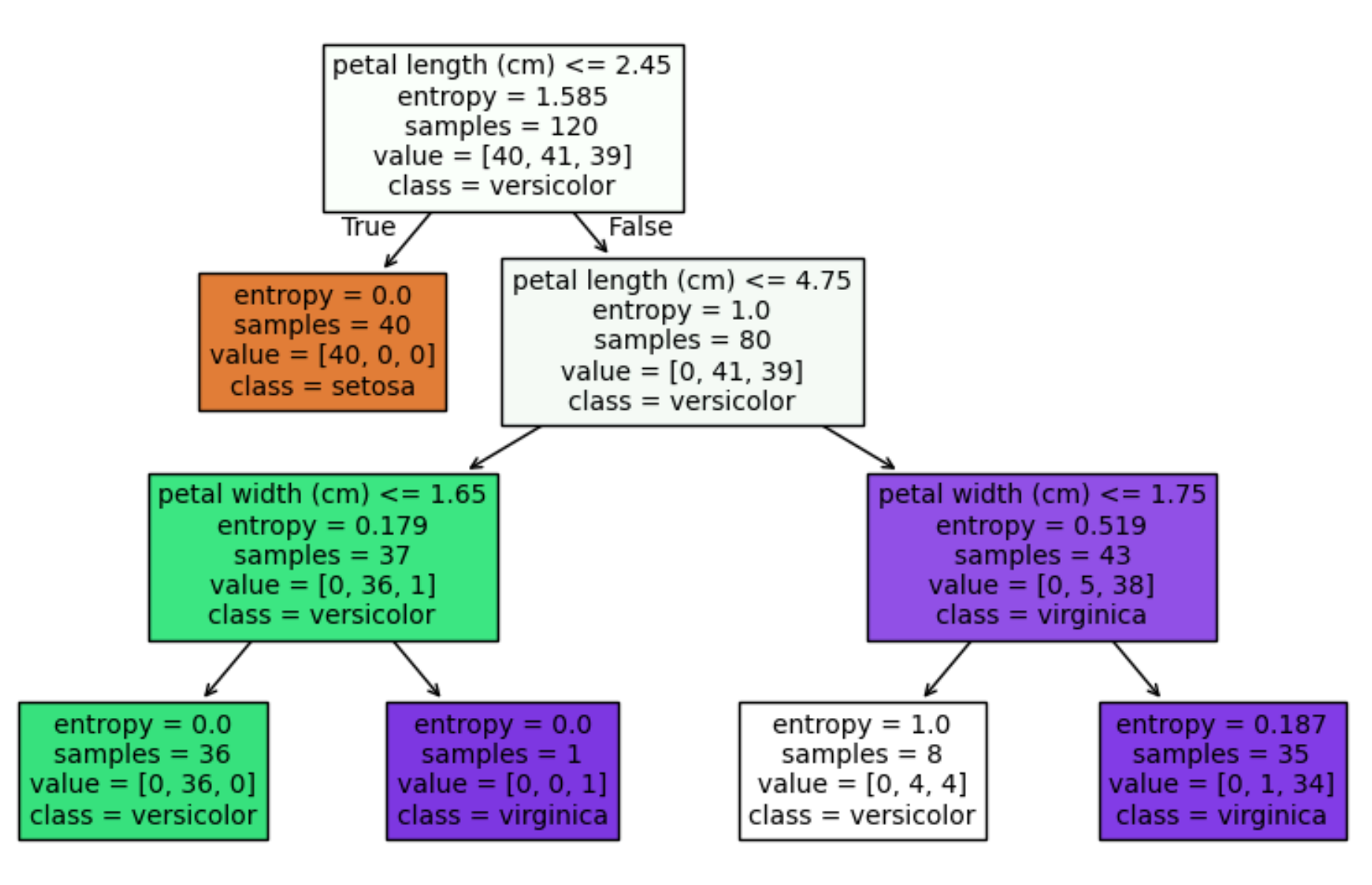
1. **Splitting Mechanism (Binary vs. Multi-way):**
   - **Manual ID3:** When we calculate manually, we often use the ID3 algorithm which creates a branch for *each* unique value of a categorical feature (multi-way split). For example, 'age' would split into three branches: `<=30`, `31-40`, and `>40`.
   - **Scikit-Learn Implementation:** The library uses an optimized version of the CART algorithm. This algorithm constructs purely **binary trees**. Even for categorical features (which are treated as numerical due to LabelEncoding), it finds a threshold to split the data into two groups (e.g., `age <= 1.5` vs `age > 1.5`).
2. **Handling of Categorical Data:**
   - We used `LabelEncoder` to transform text categories into numbers (0, 1, 2). The Decision Tree Classifier treats these as ordered numerical values. ID3 typically treats categories as nominal (unordered) values. This difference in interpretation affects how the splits are chosen and the resulting tree structure.

Therefore, while the root node might be the same (highest information gain attribute), the subsequent structure will differ because the library performs binary splits on the numerical encodings, whereas manual calculation typically performs multi-way splits on the categories.

Another example, another dataset -- Iris

In [16]:
```python
from sklearn.datasets import load_iris

# 1. Load the Iris dataset
iris = load_iris()
X = iris.data  # Features
y = iris.target  # Target labels

# 2. Split the dataset into training (80%) and testing (20%)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Create and train a Decision Tree model with entropy criterion
clf = DecisionTreeClassifier(criterion='entropy', max_depth=3, random_state=42)
clf.fit(X_train, y_train)

# 4. Make predictions on the test set
y_pred = clf.predict(X_test)

# 5. Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy:.2f}")

# 6. Visualize the Decision Tree
plt.figure(figsize=(10, 6))
plot_tree(clf, filled=True, feature_names=iris.feature_names, class_names=iris.target_names)
plt.show()
```

```
Model Accuracy: 1.00
```


petal length (cm) <= 2.45
entropy = 1.585
samples = 120
value = [40, 41, 39]
class = versicolor
True
False
entropy = 0.0
samples = 40
value = [40, 0, 0]
class = setosa
petal length (cm) <= 4.75
entropy = 1.0
samples = 80
value = [0, 41, 39]
class = versicolor
petal width (cm) <= 1.65
entropy = 0.179
samples = 37
value = [0, 36, 1]
class = versicolor
petal width (cm) <= 1.75
entropy = 0.519
samples = 43
value = [0, 5, 38]
class = virginica
entropy = 0.0
samples = 36
value = [0, 36, 0]
class = versicolor
entropy = 0.0
samples = 1
value = [0, 0, 1]
class = virginica
entropy = 1.0
samples = 8
value = [0, 4, 4]
class = versicolor
entropy = 0.187
samples = 35
value = [0, 1, 34]
class = virginica